INTERNATIONAL CONFERENCE ON RESIDUAL STRESSES

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ICRS 12

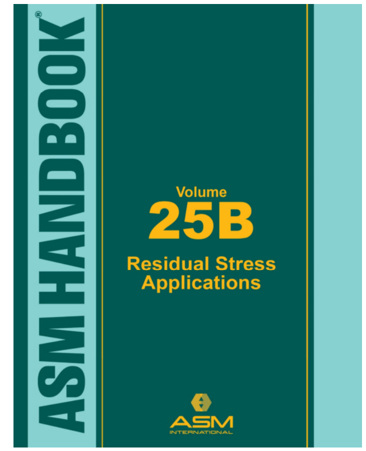
Case Studies of In-Situ Residual Stress Measurements on Bridges and Structures

James Pineault, Gabe Grodzicki, Charles Sheridan, Michael Brauss
Proto Manufacturing

ASM INTERNATIONAL

PROTO X-RAY DIFFRACTION

ASM Handbook, Volume 25B: *Residual Stress Applications*



<u>Volume Editors:</u>
David Furrer, James Pineault, Michael Hill

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<u>Divisions:</u>

- Div. 6—Modeling of Residual Stress
- Div. 7—Residual Stress Quality Control
- Div. 8—Residual Stress in Design, Production, and Sustainment
- Div. 9—Failures Due To Residual Stresses
- **<u>Div. 10—Residual Stress Case Studies by Industrial Sector</u>**

Why Talk About XRD on Bridges and Structures?

1) Dead load measurement
2) Load path determination
3) Load shedding due to corrosion
4) Validation of stress corrosion cracking
5) Validation of stress mitigation techniques (peen, HT, etc.)
6) Baseline stress measurement for enhanced structural monitoring - life prediction (i.e. live loads are known)
7) Finite element model validation of baseline
8) Supplier quality control

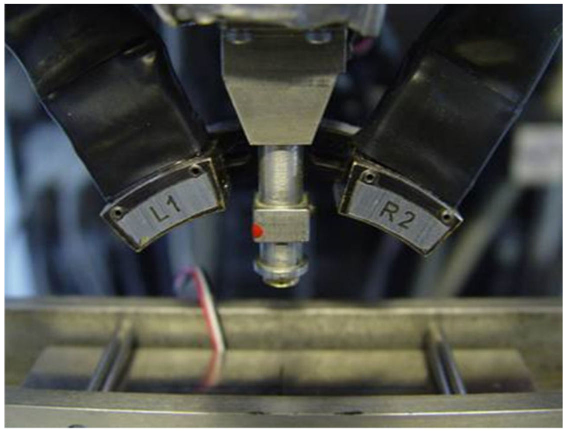
Residual Stress Measurements on the Orthotropic Deck of a Suspension Bridge

The five steps required to calibrate instrumentation and to validate stress measurement results:

1. Alignment of X-ray Diffraction Instrumentation ASTM E915
2. X-ray Elastic Constant Determination ASTM E1426
3. Surface Condition Evaluation ASTM E1426
4. Collection Parameter Selection & Optimization ASTM E2860
5. Repeatability and Reproducibility Determination

(See ASM Handbook Volume 11 – Failure Analysis & Prevention)

Segment of GWB deck shown in 4-point bend fixture for testing via ASTM E1426



Summary of GW Bridge Steel Four Point Bend Data

assumed XEC =	24,500.00 ksi
GW Bridge Steel Young's Modulus =	29,000 ksi

Sample 1	Trial 1	Trial 2	Average
Correlation Factor	1.120357278	1.157282238	1.138819758
GW Steel (h.k.l) 211 XEC (ksi)	27449	28353	27901
Error (ksi)	± 429	± 428	± 428

0.015339952

GW Bridge Steel (h.k.l) 211 XEC (ksi) = 27,901 ± 428 ksi

AST result	191004 ± 4031	MPa
AST result	27,702 ± 585	ksi

0.02110427

Form 132 - Standard Summary Report - Residual Stress — Proto Manufacturing Limited

George Washington Bridge - PS11037T - GW Bridge Steel Four Point Bend

Date: 10-Feb-14

Setup Parameters - Residual Stress:

Target: Cr ($K\alpha_{avg}$ 2.291 Angstroms)	Oscillation(s): Beta 3°
Target Power: 25 kV, 30 mA	Collection Time: 0.2 second x 20 exposures
Gain Material: β Ti	Total Collection Time: 3 minutes
Gain Power: 18 kV, 30 mA	Peak Fit: Gaussian
Filters: None	Two Peak Model: No
Material: GW Bridge Steel	LPA Correction On: Yes
Crystallographic Plane: {211}	Gain Correction: P/G
Bragg Angle (2θ): 156.31°	Background Subtraction: Linear
Aperture: 0.5 mm x 4 mm	Peak Shift Method: Absolute Peak
Powder Correction: No	Psi Zero Assignment: Curve Fit Linear

ASTM E 915:

Operator: Jeff Nantais

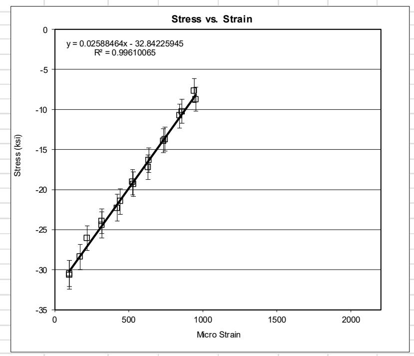
Material: Fe Loose Powder

	Stress (ksi)	Error (ksi)
1	-1.6	± 0.8
2	-1.2	± 0.8
3	-1.1	± 0.8
4	-1.3	± 0.8
5	-1.0	± 0.8
Average:	**-1.2**	**± 0.8**
Standard Deviation:	**±0.2**	

Coupon 1 - Trial 1

Micro Strain	Corrected	Stress (ksi)
98	104	-30.5 ± 1.6
207	219	-26.1 ± 1.5
304	322	-24.0 ± 1.6
419	444	-21.5 ± 1.6
500	529	-19.1 ± 1.6
600	635	-16.4 ± 1.5
704	745	-13.7 ± 1.5
812	860	-10.3 ± 1.5
902	955	-8.7 ± 1.5
894	946	-7.6 ± 1.5
800	847	-10.8 ± 1.5
696	737	-13.9 ± 1.5
595	630	-17.2 ± 1.5
503	532	-19.3 ± 1.5
398	421	-22.3 ± 1.7
303	321	-24.4 ± 1.6
163	173	-28.4 ± 1.6
98	104	-30.7 ± 1.8

Note: "-" indicates compressive residual stress'
"+" indicates tensile residual stress'



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.998048
R Square	0.996101
Adjusted R S	0.995857
Standard Err	0.473379
Observations	18

ANOVA

	df	*SS*	*MS*	*F*	*nificance F*
Regression	1	915.9038	915.904	4087.2509	1E-20
Residual	16	3.585408	0.22409		
Total	17	919.4892			

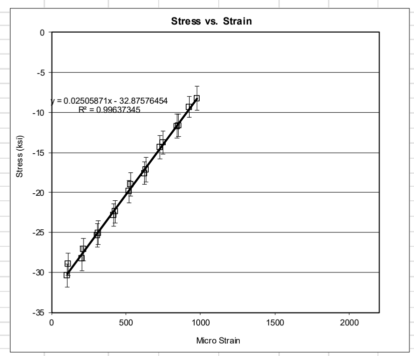
	Coefficients	*standard Err*	*t Stat*	*P-value*	*ower 95%*	*pper 95%*	*wer 95.0*	*Upper 95.0%*
Intercept	-32.8423	0.241507	-135.989	6.126E-26	-33.35	-32.33	-33.35	-32.330287
X Variable 1	0.025885	0.000405	63.9316	1.051E-20	0.025	0.027	0.025	0.026742947

error 0.015642
0.015642

Coupon 1 - Trial 2

Micro Strain	Corrected	Stress (ksi)
103	109	-29.0 ± 1.4
205	217	-27.2 ± 1.4
298	315	-25.1 ± 1.5
406	430	-22.4 ± 1.4
502	531	-19.0 ± 1.4
599	634	-17.2 ± 1.5
706	747	-13.8 ± 1.4
805	852	-11.6 ± 1.4
926	980	-8.3 ± 1.5
876	927	-9.3 ± 1.3
799	846	-11.7 ± 1.5
691	731	-14.4 ± 1.5
593	628	-17.6 ± 1.4
495	524	-19.9 ± 1.4
397	420	-22.9 ± 1.4
291	308	-25.4 ± 1.4
196	207	-28.3 ± 1.6
100	106	-30.4 ± 1.5

Note: "-" indicates compressive residual stress'
"+" indicates tensile residual stress'



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.998185
R Square	0.996373
Adjusted R S	0.996147
Standard Err	0.43891
Observations	18

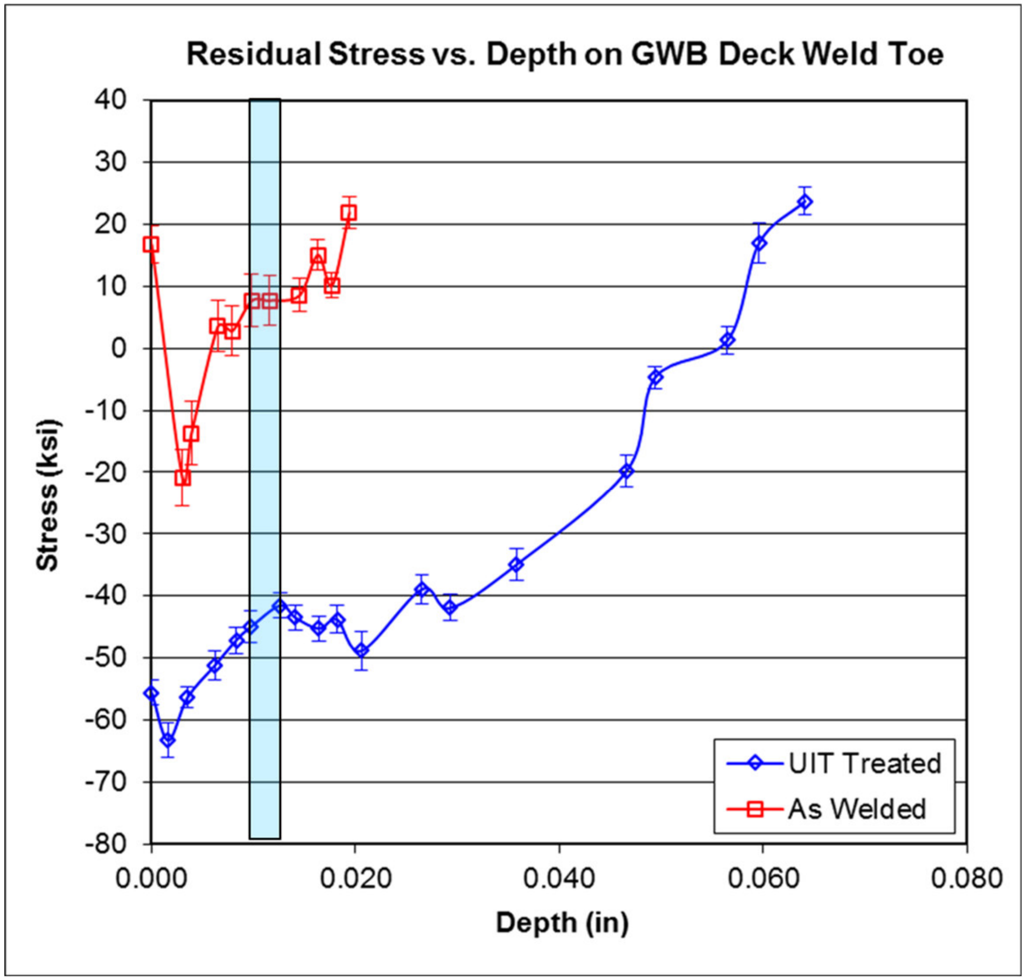
ANOVA

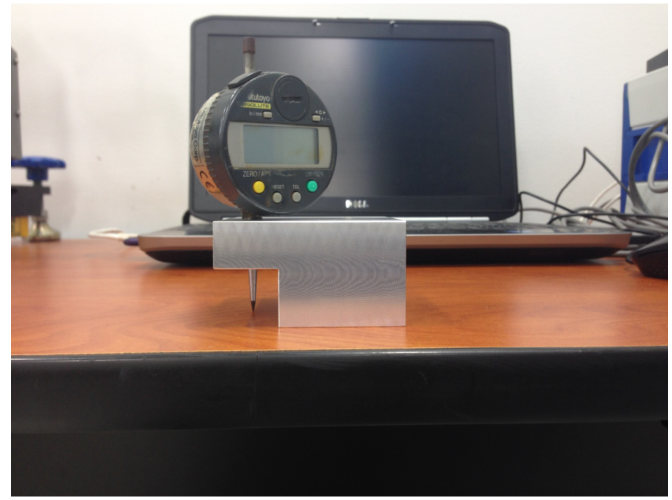
	df	*SS*	*MS*	*F*	*nificance F*
Regression	1	846.8351	846.835	4395.9104	6E-21
Residual	16	3.082265	0.19264		
Total	17	849.9174			

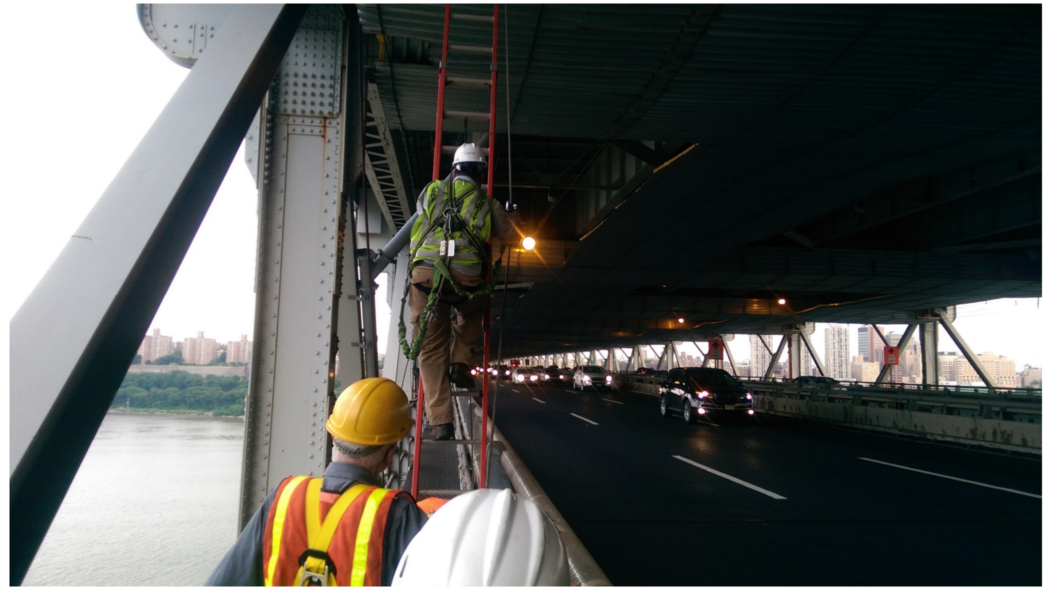
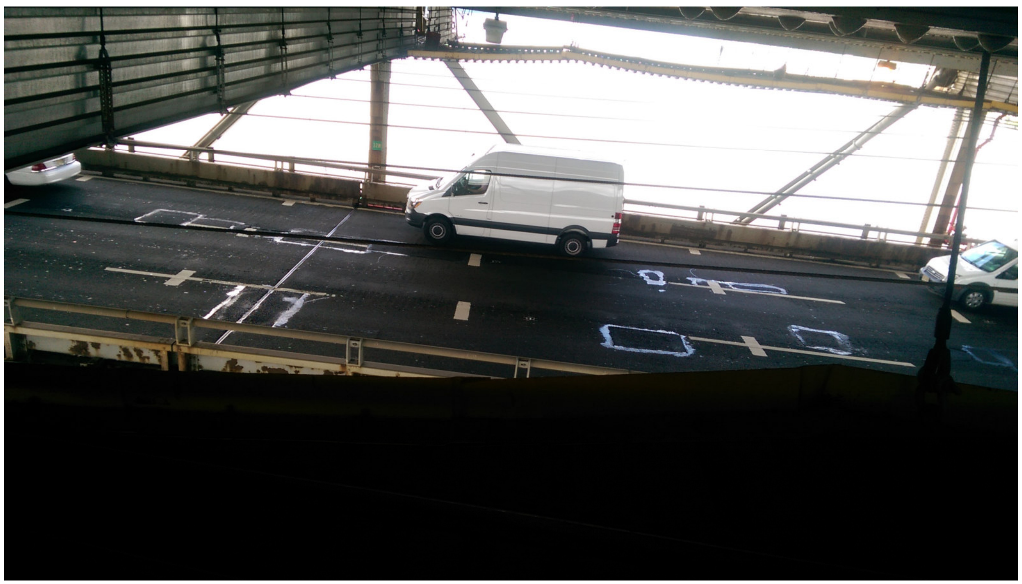
	Coefficients	*standard Err*	*t Stat*	*P-value*	*ower 95%*	*pper 95%*	*wer 95.0*	*Upper 95.0%*
Intercept	-32.8758	0.224962	-146.139	1.938E-26	-33.35	-32.4	-33.35	-32.398867
X Variable 1	0.025059	0.000378	66.3017	5.885E-21	0.024	0.026	0.0243	0.02585993

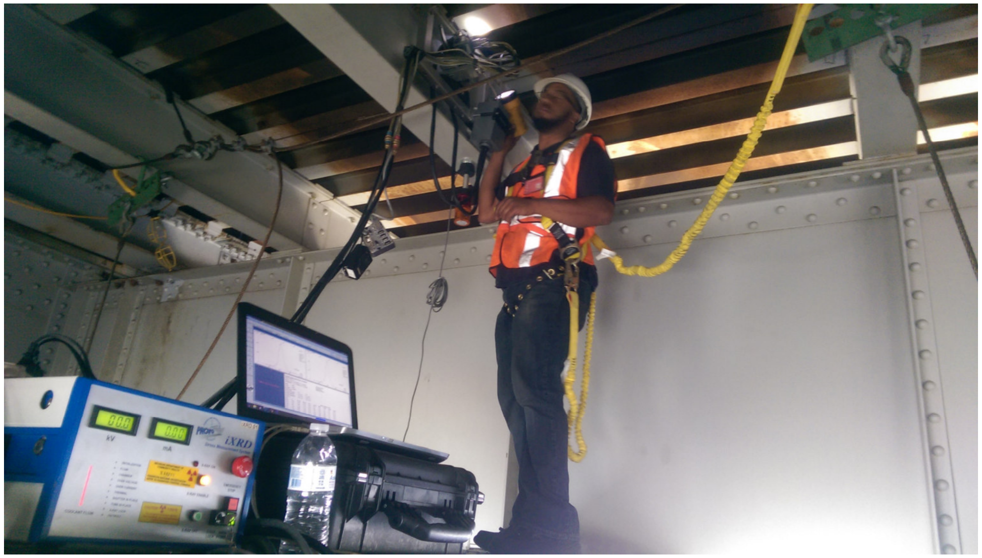
error 0.015083
0.015083

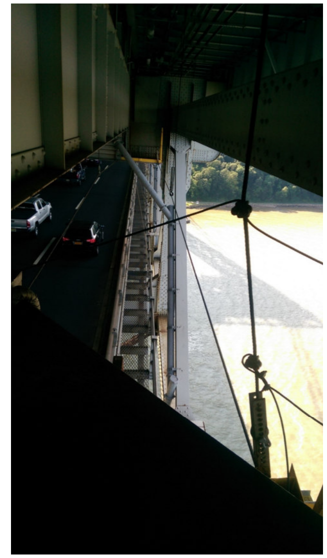
RS Profiles on Deck Welds Lab Work

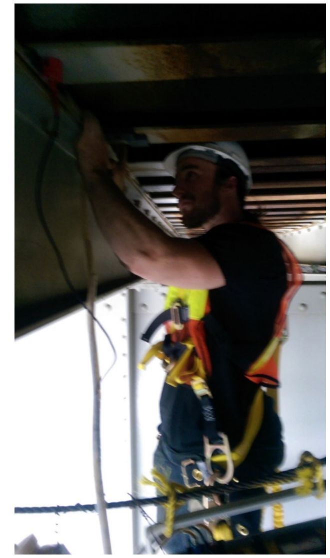






iXRD



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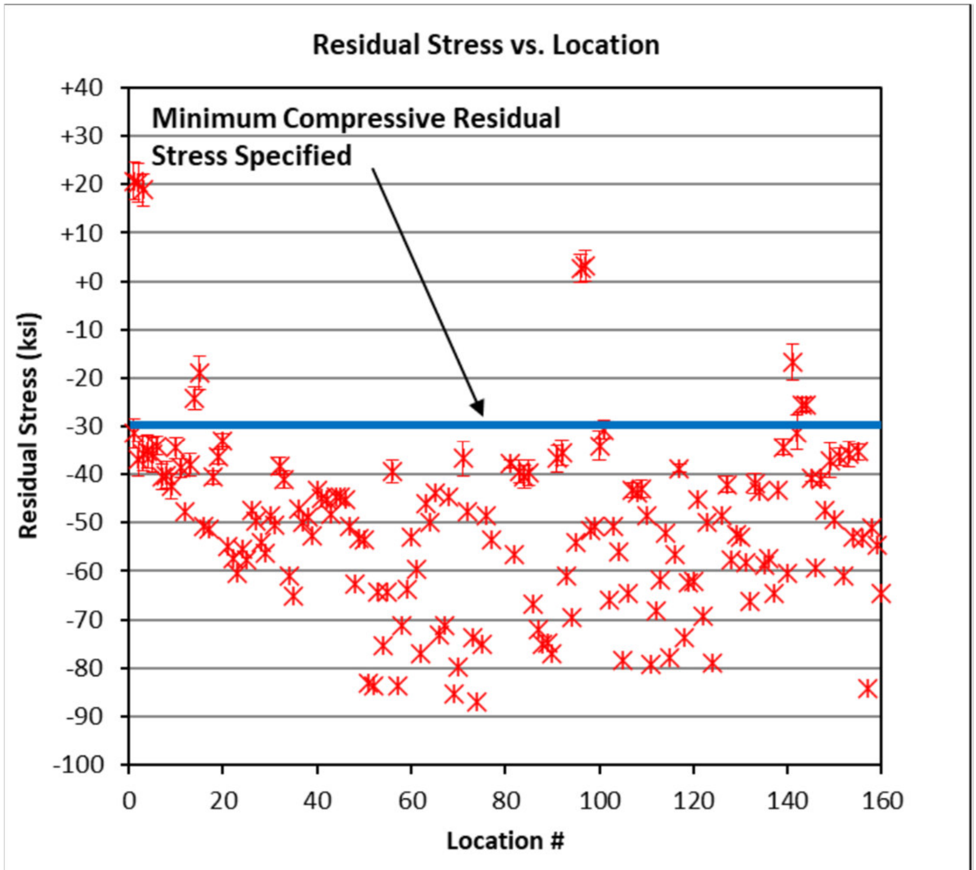
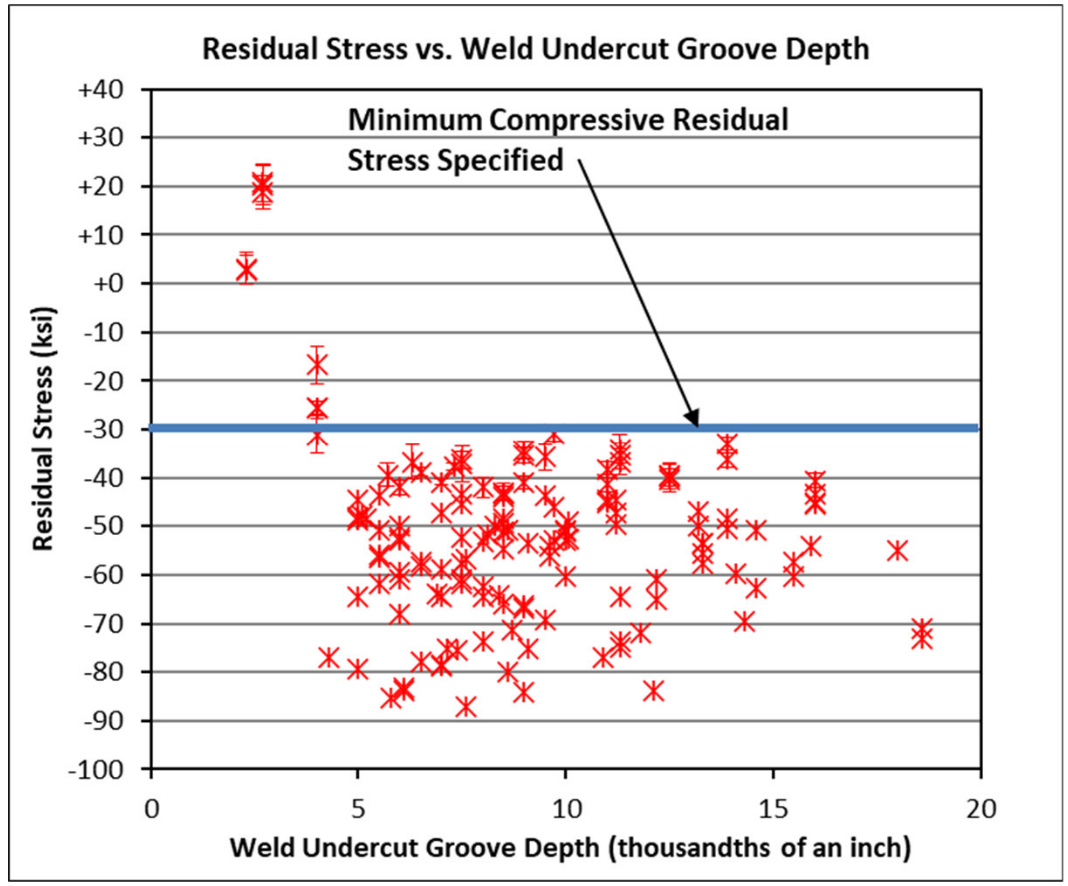
RS Analysis and Correlations





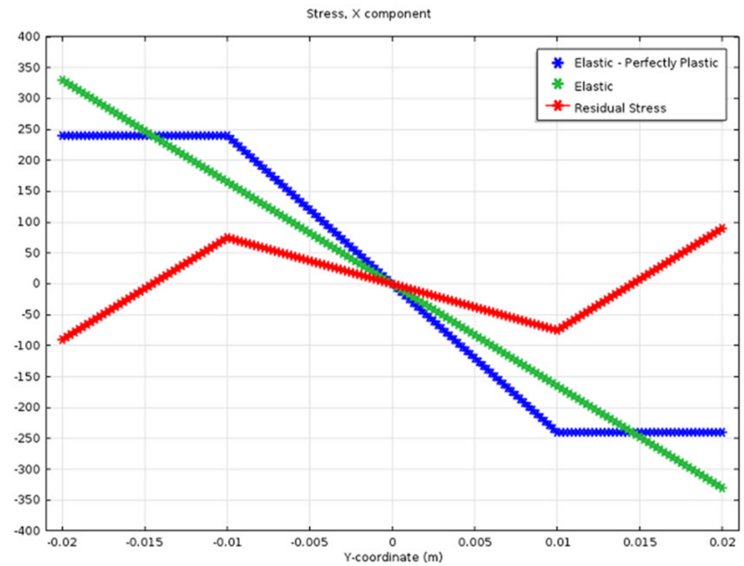
Conclusions

- XRD stress measurements were successfully performed on welds connecting structural members supporting the upper-level orthotropic deck on the GWB spanning the Hudson River between Fort Lee, NJ and Manhattan, NY.
- A high level of confidence in results requires preliminary studies
- Results provide insight into how well UIT works (i.e. depth and magnitude of compressive RS) and how treated and untreated welds can be distinguished.
- The measurement results indicate that the UIT process was successfully used to impart compressive stress in the welds on the GWB.
- XRD was used as a QA tool to audit the UIT process.

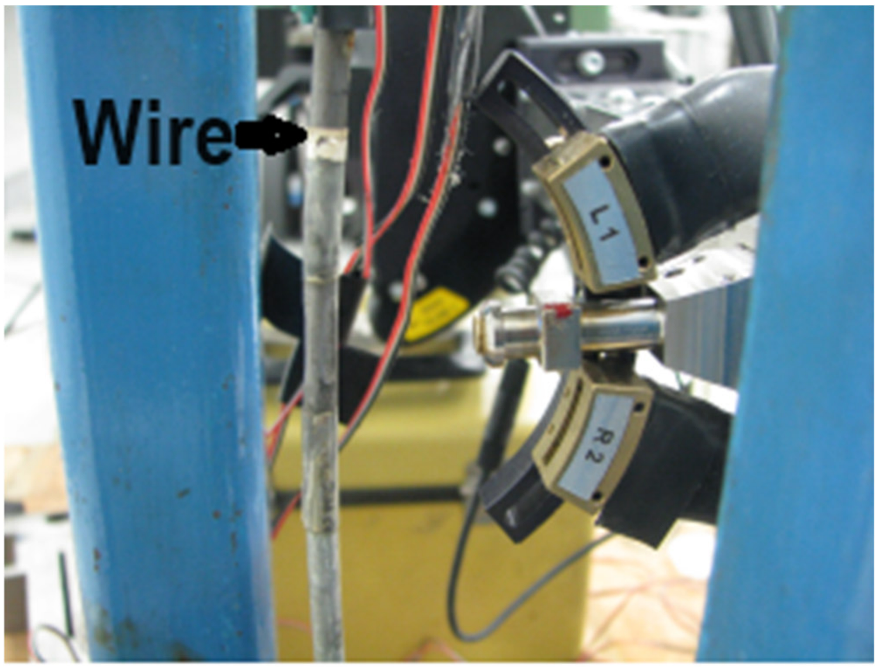
Residual Stress Measurements on Suspension Bridge Wire

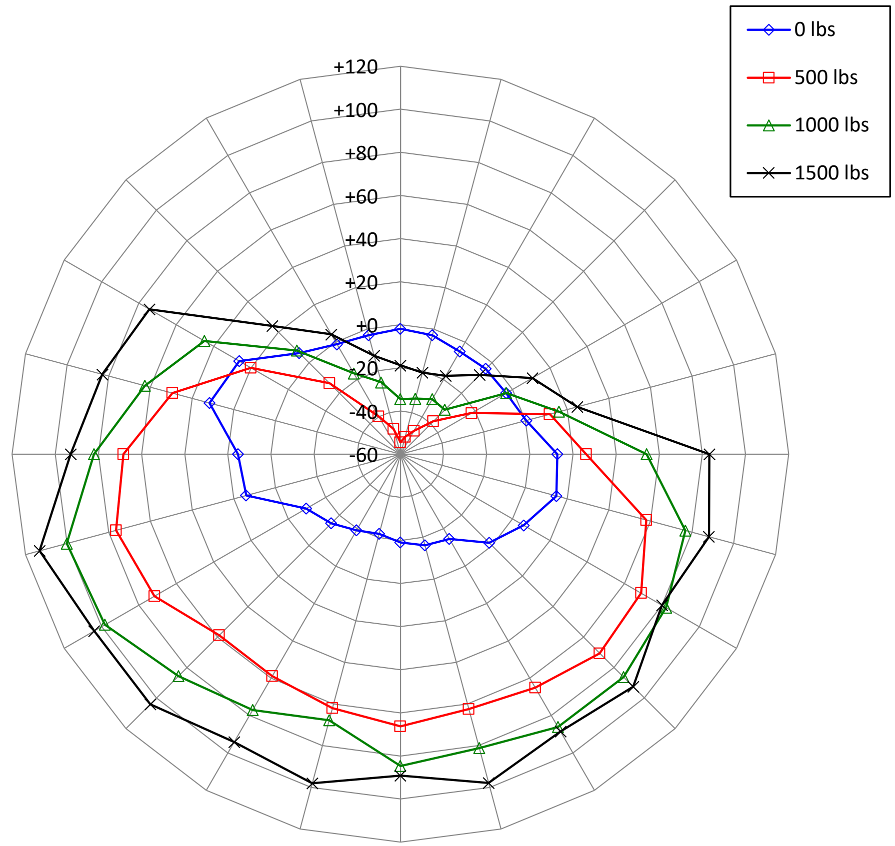
Primer on Residual Stress on Bridge Wire and Post-tensioning Tendons

- Wire manufactured by extrusion through a die have residual stresses and when the wire is coiled, additional residual stresses are imparted.
- This can be observed visually by the camber of the wire if removed.

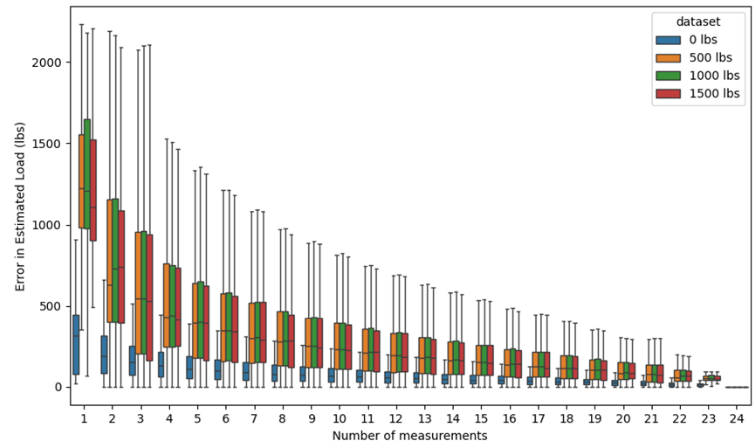


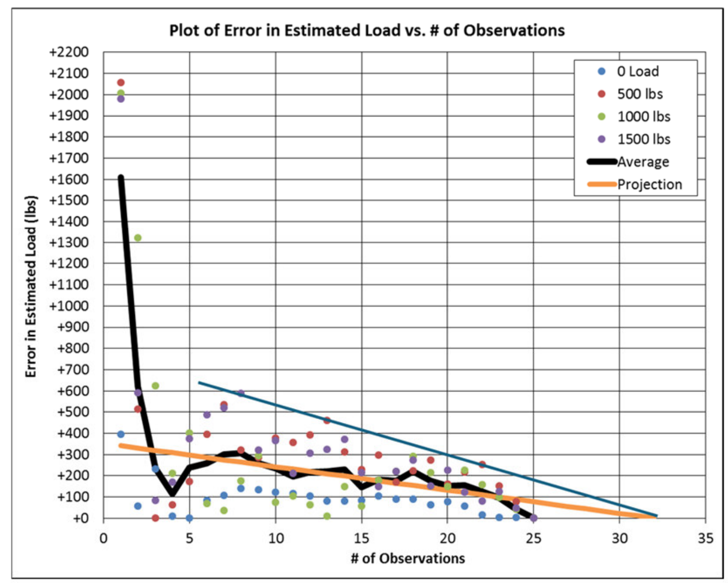
Do Your Homework: Measuring Known Dead Load on Suspension Bridge Wire



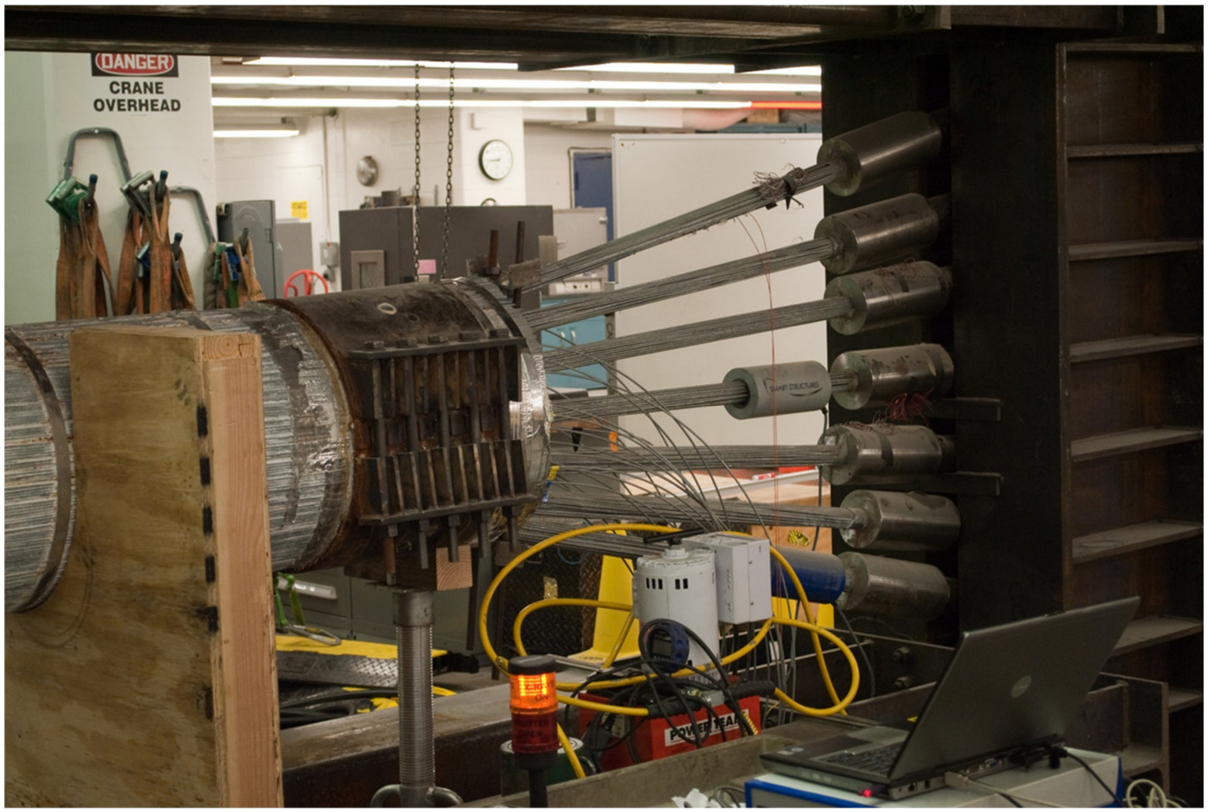


Angular Position	Total Stress (ksi) Under Various Axial Tensile Loads			
	0 lbs	500 lbs	1000 lbs	1500 lbs
0º	-2 ± 2	-55 ± 5	-34 ± 4	-19 ± 3
15º	-3 ± 2	-52 ± 4	-33 ± 3	-21 ± 3
30º	-5 ± 2	-48 ± 4	-31 ± 3	-18 ± 2
45º	-4 ± 2	-38 ± 4	-31 ± 3	-8 ± 2
60º	-3 ± 2	-22 ± 3	-3 ± 2	+11 ± 2
75º	0 ± 2	+12 ± 2	+16 ± 2	+25 ± 2
90º	+13 ± 2	+26 ± 3	+54 ± 3	+83 ± 7
105º	+15 ± 2	+58 ± 5	+77 ± 6	+88 ± 6
120º	+6 ± 2	+69 ± 4	+82 ± 6	+80 ± 6
135º	-2 ± 2	+71 ± 5	+86 ± 5	+93 ± 6
150º	-15 ± 2	+65 ± 4	+86 ± 6	+89 ± 6
165º	-16 ± 2	+62 ± 4	+81 ± 5	+98 ± 6
180º	-19 ± 2	+66 ± 3	+85 ± 3	+89 ± 3
195º	-22 ± 1	+62 ± 2	+68 ± 2	+98 ± 3
210º	-19 ± 1	+59 ± 2	+77 ± 2	+94 ± 2
225º	-15 ± 2	+59 ± 2	+86 ± 3	+104 ± 3
240º	-9 ± 2	+72 ± 3	+98 ± 2	+104 ± 3
255º	+14 ± 1	+76 ± 2	+100 ± 2	+113 ± 3
270º	+15 ± 2	+68 ± 3	+82 ± 4	+93 ± 5
285º	+32 ± 2	+50 ± 3	+63 ± 3	+83 ± 4
300º	+26 ± 1	+20 ± 3	+45 ± 2	+74 ± 3
315º	+6 ± 2	-13 ± 4	+8 ± 4	+24 ± 3
330º	-1 ± 3	-40 ± 6	-17 ± 4	+4 ± 4
345º	-3 ± 3	-48 ± 7	-26 ± 5	-13 ± 4
360º	-2 ± 2	-55 ± 5	-34 ± 4	-19 ± 3
Ave Stress (ksi)	**0 ± 2**	**+21 ± 4**	**+39 ± 3**	**+54 ± 4**
Ave. Load (lbs)	**-14 ± 56**	**+595 ± 104**	**+1117 ± 98**	**+1531 ± 107**





Full-Scale Mockup



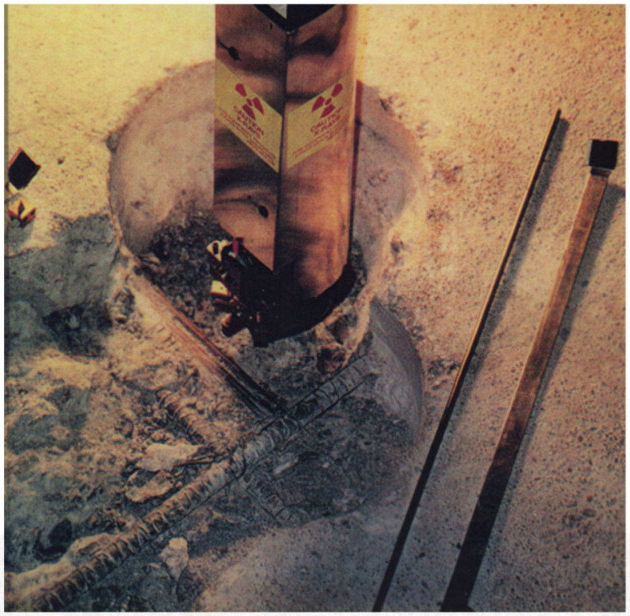
- Random wire orientations were sampled on a full-scale loading test fixture loaded up to approximately 1500 lb.
- The average stress calculated using ten observations was found to be +53.1 ± 3.5 ksi. This result is in experimental agreement with the average stress (+54.0 ± 3.8 ksi on single wire at 1500 lbs.

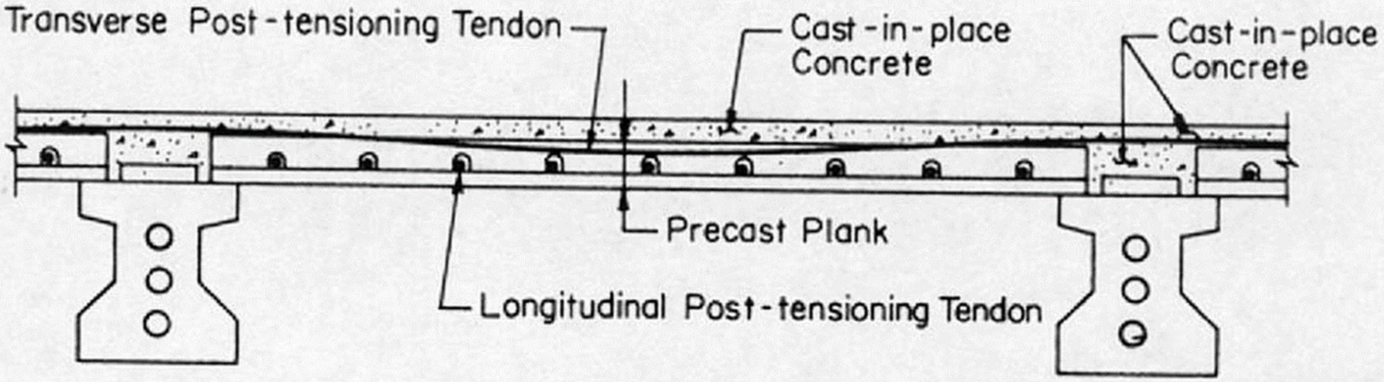
Conclusions

- XRD can be used to determine the stresses on bridge wire – identify regions where SCC is most likely to occurs.
- If well understood, the residual stress in “as manufactured” wire can potentially be fine tuned to minimize this effect.
- With a sufficient number of wires sampled, the precise load each wire bundle is carrying can be determined.

Residual Stress Measurements in Post-Tensioning Tendons

Measuring Dead Load on Post-tensioning Tendons





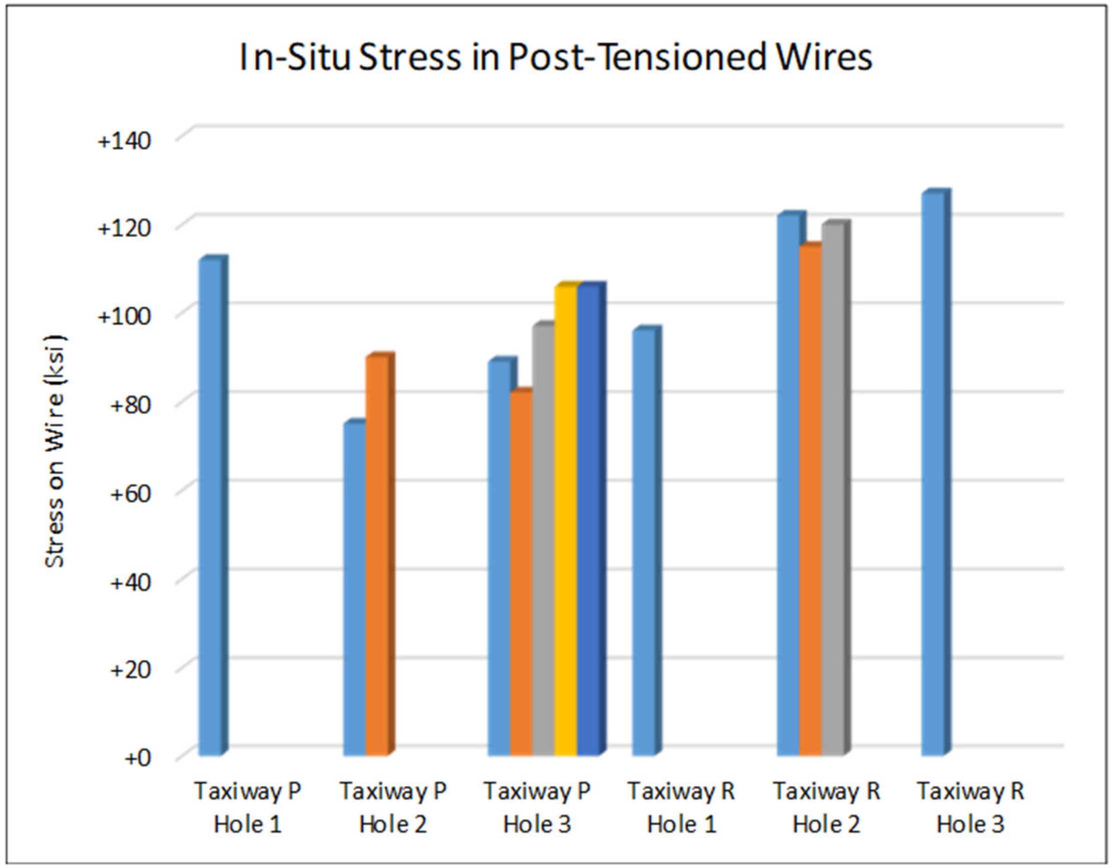


Figure 15 - Stress measurement results on installed post-tensioned tendon wires.

Figure 16 – XRD stress measurements in post-tensioned wire bundle in a structure.

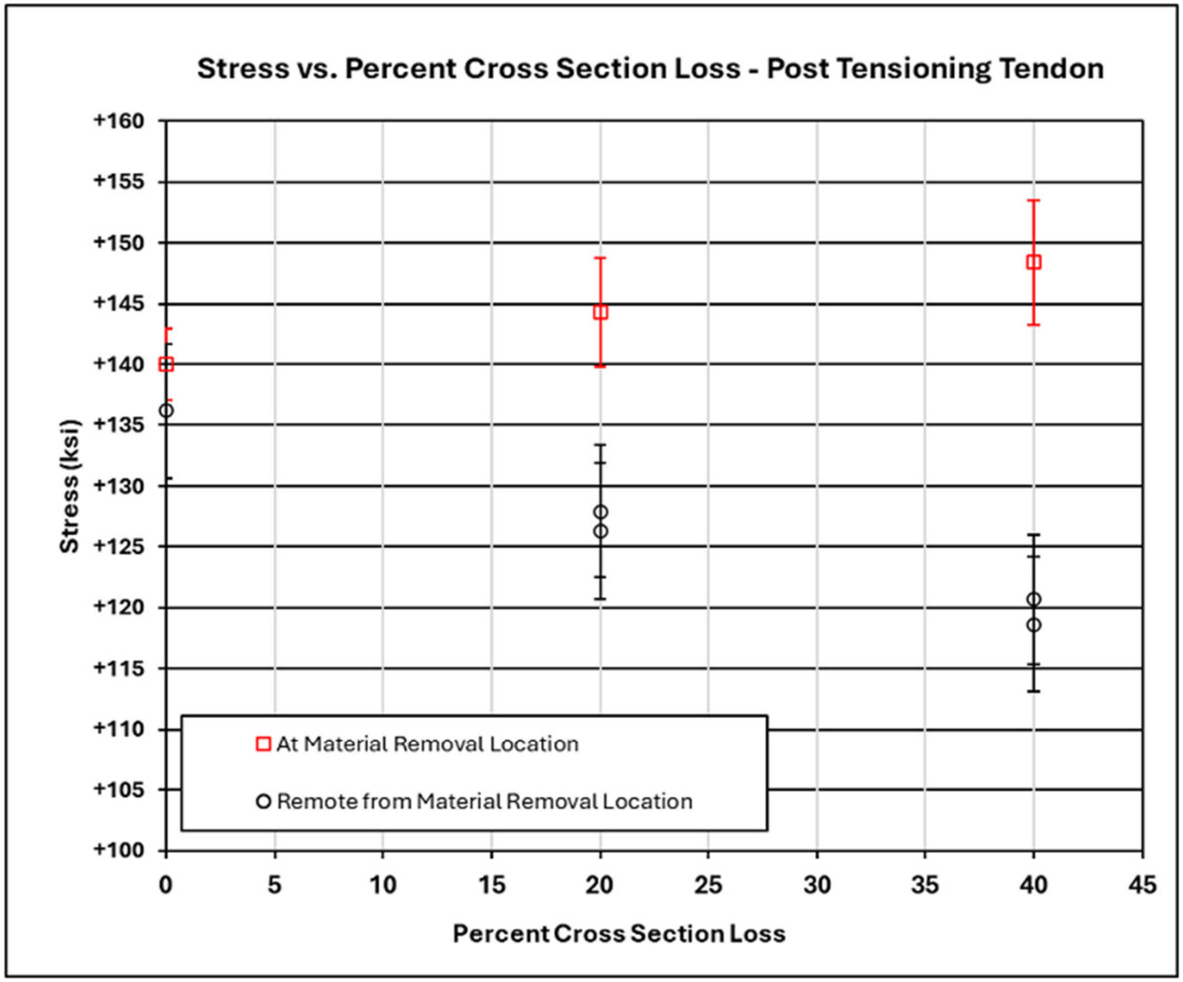


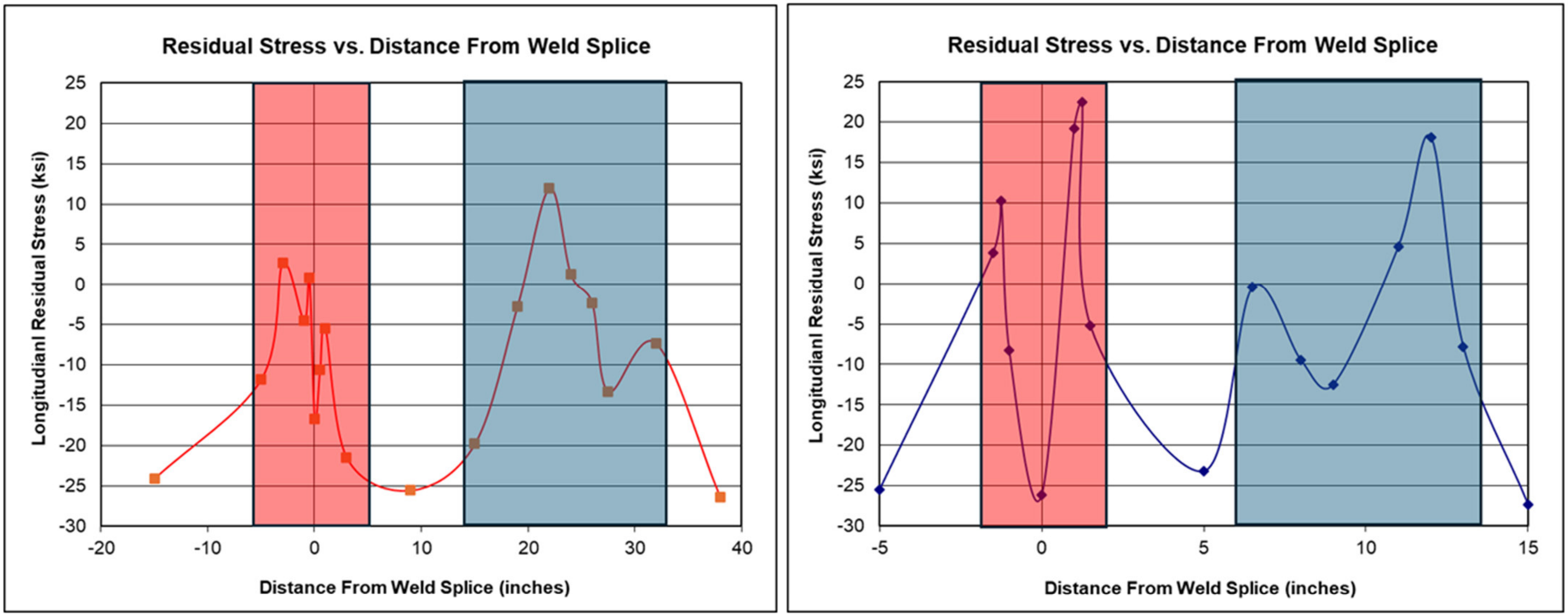
Figure 17 – Plot of stress vs. percent cross section loss.

Conclusions

- The stresses found ranged from +517 MPa to +772 MPa (+75 ksi to +112 ksi) with an average of +652 MPa (+95 ksi) in tension on Taxiway P.
- The stresses found ranged from +662 MPa to +876 MPa (+96 ksi to +127 ksi) with an average of +800 MPa (or +116 ksi) in tension.
- These results are within 20% of the initial estimates at the time of installation i.e. it was estimated that the wires should have tensile stresses in the range of 703 MPa (+102 ksi) to +986 MPa (+143 ksi) as compared to the rated yield strength of approximately 225 ksi (~1500 MPa).
- Depth profile on an individual wire shows how it sheds load to the other wires in the bundle.

Residual Stress Measurements on Steel Box Girder Monorail Structure

Figure 10 – Field portable XRD system measuring RS on monorail box girder structure.



a) b)

Figure 11 – Residual stress across a) top side weld splice and adjacent heat straightened region, and b) bottom side weld splice and adjacent heat straightened region on monorail box girder. (Note: Red colored areas represent weld and weld HAZ regions, Blue colored areas represent Heat Straightened regions)

Conclusions

- XRD was successfully used to measure the residual stresses in welded and heat straightened regions of a monorail box girder that could not easily or economically be modeled.
- Using the obtained data, engineers were able to fine tune fatigue life estimates of the structure, and from these estimates, more robustly establish suitable maintenance inspection intervals so that fatigue cracks can be detected and repaired long before they become a structural integrity issue.

Main Takeaways:

1) XRD can be was used as a QA tool to audit stress management/engineered residual stress processes such as UIT.

2) XRD can be used to determine the stresses on bridge wire – identify regions where SCC is most likely to occurs and assist in improved manufacturing methods.

3) With a sufficient number of wires sampled, the precise dead load each wire bundle is carrying can be determined.

4) Post-tensioning tendons can be characterized to determine if sufficient tension remains after years of in-service usage.

5) Depth profiles on individual wires can show load is shed.

6) Measurements on individual structural members can show load path/distribution of load.

7) XRD can be used to measure the residual stresses in welded and heat straightened regions of structural members that cannot easily be modeled.

8) Information obtained can help engineers fine tune fatigue life estimates of the structure and more robustly establish suitable maintenance inspection intervals so that fatigue cracks can be detected and repaired long before they become a structural integrity issue.

Questions?